

Name:

Enrolment No:



UPES

End Semester Examination, December 2025

Course: Advanced Engineering Mathematics-I

Program: B.Tech. (SoCS)

Course Code: MATH1059

Semester: I

Time : 03 hrs.

Max. Marks: 100

Instructions: Attempt all the questions. There are internal choices in the question number 9 and 11. Calculator is allowed.

SECTION A

(5Qx4M=20Marks)

S. No.		Marks	CO
Q 1	Discuss the maxima and minima of $x^2 + y^2 + 6x + 12$.	4	CO1
Q 2	Determine the value of $\Gamma\left(-\frac{5}{2}\right)$.	4	CO2
Q 3	Change the order of integration for $\int_{-a}^a \int_{\frac{y^2}{a}}^{\frac{y^2}{a}} f(x, y) dx dy.$	4	CO2
Q 4	Compute $\text{div}(\text{grad } \phi)$, for the scalar valued function $\phi(x, y, z)$.	4	CO3
Q 5	Find all the possible values of a and b , for which given differential equation is exact $(y + x^3)dx + (ax + by^3)dy = 0.$	4	CO4

SECTION B

(4Qx10M= 40 Marks)

Q 6	Use Mean value theorem to show that for any $x \in \left(0, \frac{\pi}{2}\right)$, the ratio $\frac{\sin x}{x}$ satisfies the inequality $\frac{2}{\pi} < \frac{\sin x}{x} < 1.$	10	CO1
Q 7	Evaluate the double integral using polar coordinates (r, θ) $\int_0^1 \int_x^{\sqrt{2x-x^2}} (x^2 + y^2) dx dy.$	10	CO2
Q 8	A competitive interaction is described by the Lotka-Volterra competition model	10	CO5